

GovConnect — Government Scheme Recommendation App (Product Workspace)



GovConnect (working title) is a Flutter-based Government Scheme Recommendation App for India that recommends only the Central & State welfare schemes a citizen is *actually eligible for* — replacing hours of portal-hopping with a single personalized feed. This workspace is the single source of truth for product, UX, and engineering.

Doc owner: Hepin Dobariya · **Status:** Draft v1.0 · **Stack:** Flutter + Firebase · **Region:** India-first



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1. Executive Summary

The Problem. India runs *hundreds* of Central and State welfare schemes across scholarships, pensions, farming subsidies, housing, healthcare, and business support. Yet most eligible citizens never claim their benefits — not because they don't qualify, but because they never discover the schemes, can't decode the eligibility rules, or can't navigate fragmented government portals.

Target Audience. Students, senior citizens, farmers, women, job seekers, MSME owners, disabled citizens, and low-income families — with a strong focus on rural, elderly, and first-time smartphone users.

The Solution. GovConnect asks a citizen a short, plain-language eligibility questionnaire (age, income, gender, occupation, state, category, etc.) and uses a **rule-based recommendation engine** to surface *only* the schemes they qualify for. Each scheme card explains benefits, documents required, and a direct link to the official application — in the user's own language.

Market Gap. Existing government portals (MyScheme, state portals) are *information repositories*, not *personalized advisors*. They are search-heavy, English/Hindi-dominant, and assume the user already knows what to look for. GovConnect flips this: it tells you what you're eligible for before you even ask.

Vision. "Every Indian citizen discovers and claims every benefit they are entitled to — in their language, in under 5 minutes." GovConnect aims to become the default benefits discovery layer for India, eventually assisting with document readiness and application tracking.

2. Problem Statement

Access to welfare in India is limited less by *eligibility* and more by *discoverability* and *usability*.

#	Problem	Why it matters
1	India has hundreds of welfare schemes across Central + 28 States + 8 UTs	Sheer volume makes manual discovery impossible for a citizen
2	Citizens don't know which schemes they qualify for	Benefits go unclaimed; leakage of intended welfare
3	Portals are information-heavy, not personalized	Users must self-diagnose eligibility from dense legal text
4	Rural & elderly users struggle to navigate	Low digital literacy + English/Hindi-only UIs exclude them
5	Users waste time across multiple websites	Central, state, and departmental portals are fragmented

 **Context (illustrative — verify before publishing):** India has 900M+ internet users and 750M+ smartphone users, yet a large share of eligible beneficiaries remain unaware of schemes meant for them. Awareness — not eligibility — is the primary barrier. *(Replace with cited figures from MyScheme / NITI Aayog / NSSO before any external use.)*

Consequence: Welfare that is *funded and available* never reaches the intended citizen. GovConnect closes the **awareness-to-application gap**.

3. Objectives

Functional Objectives

- ☐ Recommend schemes based on a rule-based eligibility engine
- ☐ Support Central + State scheme datasets with structured eligibility criteria
- ☐ Provide a short, plain-language eligibility questionnaire
- ☐ Enable search, filtering, and bookmarking of schemes
- ☐ Deliver scheme detail pages with benefits, documents, and official apply links
- ☐ Support multilingual UI (regional languages)
- ☐ Send deadline & new-scheme notifications

Business Objectives

- ☐ Achieve high scheme-discovery-to-official-site conversion
- ☐ Build a maintainable, verifiable scheme database as a defensible asset
- ☐ Establish partnerships with government bodies, NGOs, and CSR sponsors
- ☐ Reach sustainable monetization without compromising free citizen access

Social Objectives

- ☐ Increase welfare scheme awareness among underserved groups
 - ☐ Reduce welfare leakage caused by lack of information
 - ☐ Make benefits accessible to low-literacy, elderly, and rural users
 - ☐ Promote digital and financial inclusion
-

4. Target Users

Segment	😞 Pain Points	🎯 Goals	📱 App Usage
Students	Don't know which scholarships apply; miss deadlines	Find & apply to scholarships, education loans	Eligibility quiz → scholarship feed → deadline alerts
Senior Citizens	Complex portals; unaware of pension schemes	Access pensions, healthcare, senior benefits	Simple guided flow, large fonts, voice assist
Farmers	Miss subsidies, PM-KISAN, crop insurance	Discover farming subsidies & insurance	State + category filter, regional language
Women	Unaware of maternity, safety, self-help schemes	Access women-focused welfare & loans	Gender-based recommendations, alerts
Job Seekers	Don't know skill/employment schemes	Find skilling, stipend, employment schemes	Occupation-based feed, apply links
MSME Owners	Miss business loans, subsidies, credit schemes	Discover MSME/business support	Business-owner filter, document checklist
Disabled Citizens	Hard to find disability-specific benefits	Access disability pensions, aids, reservations	Disability flag → tailored feed, accessible UI
Low-income Families	Unaware of ration, housing, health schemes	Access food security, PMAY, Ayushman	Income-based feed, offline access
General Citizens	Overwhelmed by scattered info	One place for all eligible schemes	Personalized home feed, search

5. 👤👤 User Personas

Persona 1 — Ramesh (Farmer)

Attribute	Detail
Age	48
Occupation	Small-scale farmer (2 acres)
Income	₹1.2 LPA
Tech Literacy	Low — uses WhatsApp & YouTube only
Goals	Get PM-KISAN, crop insurance, fertilizer subsidy
Frustrations	Can't read English portals; relies on agents who charge fees

Persona 2 — Priya (Student)

Attribute	Detail
Age	19
Occupation	Undergraduate student
Income	Family income ₹3 LPA
Tech Literacy	High — smartphone-native
Goals	Find scholarships, never miss deadlines
Frustrations	Too many portals; unclear eligibility; missed a scholarship deadline last year

Persona 3 — Lakshmi (Senior Citizen)

Attribute	Detail
Age	67
Occupation	Retired homemaker
Income	₹0 (dependent)
Tech Literacy	Very low — needs help operating phone
Goals	Old-age pension, free healthcare (Ayushman)
Frustrations	Small fonts, English UI, complex steps

Persona 4 — Anil (MSME Owner)

Attribute	Detail
Age	35
Occupation	Owns a small manufacturing unit
Income	₹8 LPA
Tech Literacy	Medium-high
Goals	Access MUDRA/CGTMSE loans, subsidies
Frustrations	Doesn't know which credit schemes apply; paperwork confusion

Persona 5 — Sunita (Woman / Job Seeker)

Attribute	Detail
Age	27
Occupation	Unemployed graduate, seeking work
Income	₹0 personal
Tech Literacy	Medium
Goals	Skilling programs, women's self-employment loans, stipends
Frustrations	Overwhelmed by scattered info; unsure what she qualifies for

6. User Journey

```

Splash Screen
↓
Language Selection  (English / Hindi / regional)
↓
Login / Guest      (phone OTP or continue as guest)
↓
Eligibility Questionnaire  (age, income, gender, state, occupation, category...)
↓
Recommendation Engine  (rule-based match → eligible schemes)
↓
Scheme Details      (benefits, documents, eligibility, deadlines)
↓
Bookmark            (save for later)
↓
Official Website    (deep link to gov portal)
↓
Apply               (user applies on official portal)

```

Journey notes

- **Guest mode** lets users get recommendations without signup (reduces drop-off); bookmarks/notifications prompt for login.
- The questionnaire is **progressive** — each answer narrows results; users can edit answers anytime from Profile.
- GovConnect **does not submit applications itself** (v1) — it routes users to the official portal to maintain trust and legal safety.

7. ✨ Features

MVP Features

- ☐ Language selection (English, Hindi + 1–2 regional to start)
- ☐ Phone OTP login + Guest mode
- ☐ Eligibility questionnaire
- ☐ Rule-based recommendation engine
- ☐ Scheme feed (personalized home)
- ☐ Scheme search + filters
- ☐ Scheme detail page (benefits, docs, apply link)
- ☐ Bookmarks
- ☐ Basic push notifications (deadlines, new schemes)
- ☐ Profile with editable eligibility answers
- ☐ Admin panel to manage scheme data

Future Features

- ☐ Full multilingual coverage (10+ languages)
- ☐ Deadline reminder engine with calendar sync
- ☐ Document checklist per scheme with readiness tracking
- ☐ Application status tracking
- ☐ Family profile (manage benefits for multiple members)
- ☐ In-app scheme comparison

Nice-to-Have Features

- ☐ AI chatbot assistant
- ☐ Voice-based navigation for low-literacy users
- ☐ OCR document checker
- ☐ Offline mode with cached schemes
- ☐ Share scheme via WhatsApp
- ☐ Dark mode

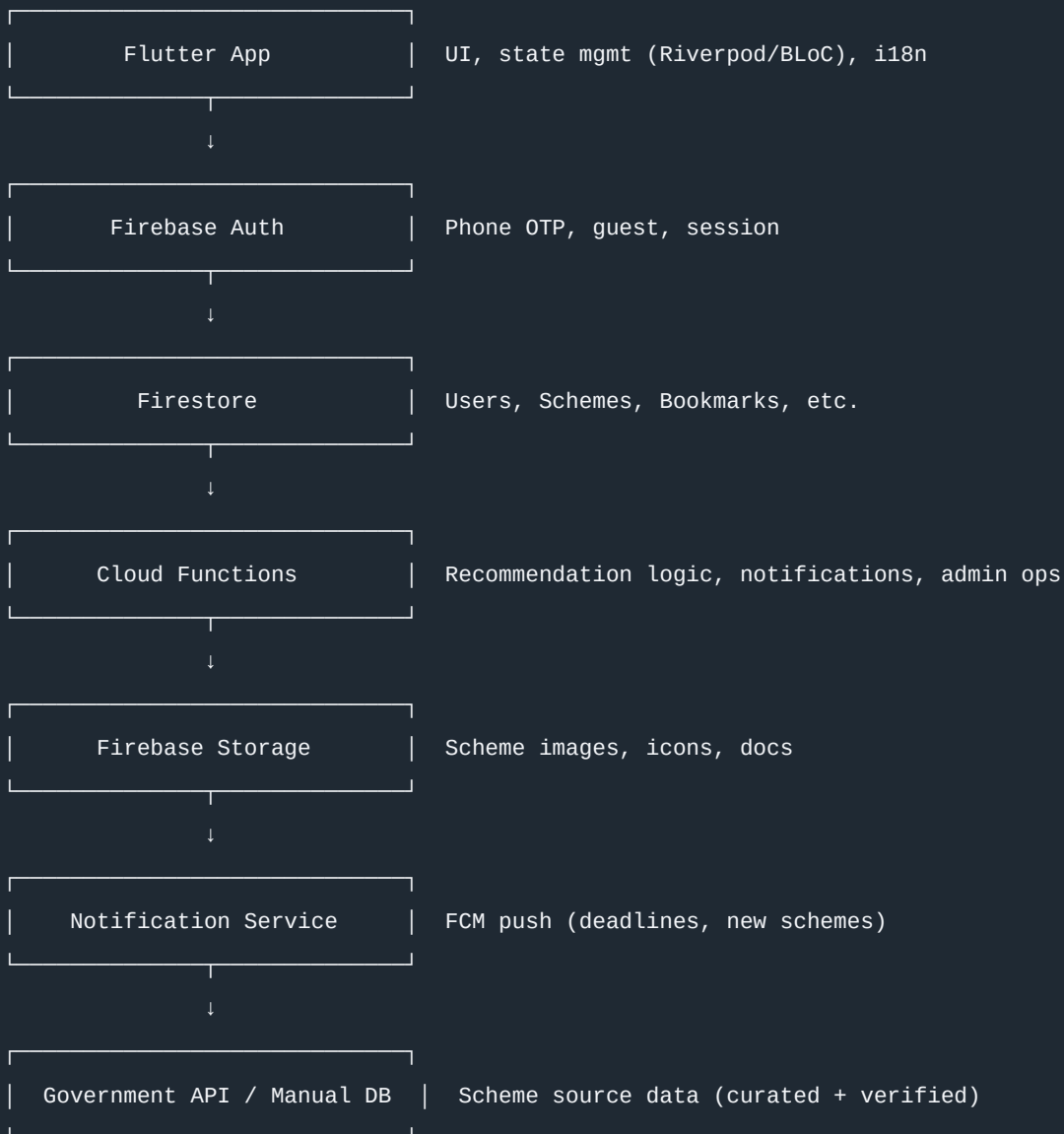
8. 📋 Functional Requirements

ID	Requirement	Description	Priority
FR-001	User Registration & Auth	Phone OTP login via Firebase Auth; guest mode; session persistence	High
FR-002	Language Selection	Select UI language at first launch; changeable from settings	High
FR-003	Eligibility Questionnaire	Collect age, income, gender, state, occupation, category, education, disability, minority, farmer, business-owner flags	High
FR-004	Recommendation Engine	Match user profile against scheme eligibility rules; return only eligible schemes ranked by relevance	High
FR-005	Scheme Search	Keyword search across scheme name, category, description	High
FR-006	Filters	Filter by state, category, benefit type, target group	High
FR-007	Scheme Detail	Show benefits, eligibility, documents, deadline, official link	High
FR-008	Bookmark	Save/unsave schemes; view bookmarked list (login required)	Medium
FR-009	Notifications	Push alerts for new schemes & approaching deadlines	Medium
FR-010	Profile Management	View/edit eligibility answers; language & notification prefs	High
FR-011	Feedback	Submit feedback / report incorrect scheme data	Medium
FR-012	Admin Panel	CRUD schemes, states, categories; manage eligibility rules; review feedback	High
FR-013	Official Apply Redirect	Deep-link/open official government portal in browser	High
FR-014	Offline Cache	Cache last-fetched schemes for offline viewing	Low

9. Non-Functional Requirements

Category	Requirement
 Performance	Home feed & recommendations load < 2s on 3G/4G; questionnaire results in < 1s; lazy-load scheme lists
 Security	Firebase Auth; Firestore security rules; encrypted data in transit (HTTPS); minimal PII stored; consent for sensitive fields (caste, disability); comply with India DPDP Act
 Accessibility	WCAG AA contrast; scalable fonts; screen-reader labels; large tap targets; voice support (future); simple guided flows for low-literacy users
 Offline Support	Cache schemes & bookmarks locally; graceful offline messaging; sync on reconnect
 Scalability	Firestore + Cloud Functions auto-scale; paginated queries; indexed scheme collection to handle 1000s of schemes & high read volume
 Localization	Full i18n framework; RTL-safe; regional language content for scheme data & UI strings
 Maintainability	Clean architecture, modular Flutter code, documented scheme data schema
 Availability	Target 99.5% uptime; graceful degradation when gov links are down

10. App Architecture



Architecture decisions

- **Client:** Flutter (single codebase, Android-first for India), Riverpod or BLoC for state, `intl` / `easy_localization` for i18n.
- **Recommendation:** Runs as a **Cloud Function** (or client-side for simple rules in MVP) so rules can change without app updates.
- **Scheme data:** Curated **manual database** in v1 (government APIs are inconsistent); designed to swap in official APIs later.
- **Notifications:** FCM triggered by scheduled Cloud Functions checking deadlines.

11. Database Design (Firestore Collections)

`users`

Field	Type	Description
uid	string	Firebase Auth UID
phone	string	Phone number
name	string	Optional display name
language	string	Preferred language code
age	number	Age
gender	string	male / female / other
income	number	Annual income (₹)
state	string	State code
occupation	string	Occupation category
category	string	General / OBC / SC / ST
education	string	Education level
isDisabled	boolean	Disability flag
isMinority	boolean	Minority flag
isFarmer	boolean	Farmer flag
isBusinessOwner	boolean	Business-owner flag
createdAt	timestamp	Created
updatedAt	timestamp	Last profile update

schemes

Field	Type	Description
<code>schemeId</code>	string	Unique ID
<code>title</code>	map	Localized name
<code>description</code>	map	Localized summary
<code>category</code>	string	Ref to categories
<code>level</code>	string	central / state
<code>state</code>	string	State code (null if central)
<code>benefits</code>	string	Benefit description
<code>documents</code>	array	Required documents
<code>eligibility</code>	map	Rule object (see §12)
<code>officialUrl</code>	string	Apply/official link
<code>deadline</code>	timestamp	Application deadline (nullable)
<code>iconUrl</code>	string	Storage image URL
<code>isActive</code>	boolean	Visible or archived
<code>updatedAt</code>	timestamp	Last verified/updated

bookmarks

Field	Type	Description
<code>bookmarkId</code>	string	Unique ID
<code>uid</code>	string	Owner
<code>schemeId</code>	string	Ref to scheme
<code>createdAt</code>	timestamp	Saved at

notifications

Field	Type	Description
<code>notificationId</code>	string	Unique ID
<code>uid</code>	string	Target user (null = broadcast)
<code>title</code>	string	Notification title
<code>body</code>	string	Message
<code>type</code>	string	deadline / new_scheme / general
<code>schemeId</code>	string	Related scheme (nullable)
<code>isRead</code>	boolean	Read state
<code>createdAt</code>	timestamp	Created

feedback

Field	Type	Description
<code>feedbackId</code>	string	Unique ID
<code>uid</code>	string	Author (nullable for guest)
<code>schemeId</code>	string	Related scheme (nullable)
<code>type</code>	string	bug / data_error / suggestion
<code>message</code>	string	Feedback text
<code>status</code>	string	open / reviewed / resolved
<code>createdAt</code>	timestamp	Created

states

Field	Type	Description
<code>stateCode</code>	string	e.g. GJ, MH
<code>name</code>	map	Localized state name
<code>isActive</code>	boolean	Active flag

categories

Field	Type	Description
categoryId	string	Unique ID
name	map	Localized name (e.g. Education, Agriculture)
iconUrl	string	Icon
order	number	Display order

12. Recommendation Algorithm (Rule-Based Filtering)

GovConnect uses **deterministic rule-based matching** — transparent, explainable, and easy to audit (critical for government data trust).

How it works

1. Build a **user profile object** from questionnaire answers.
2. Each scheme stores an **eligibility rule object** with constraints.
3. The engine evaluates every active scheme against the profile; a scheme is **eligible only if ALL its rules pass**.
4. Eligible schemes are ranked (state match, deadline proximity, category relevance).

Eligibility criteria evaluated

Criterion	Rule Example
Income	<code>income <= 250000</code>
Age	<code>age >= 18 && age <= 35</code>
Gender	<code>gender == 'female'</code>
Occupation	<code>occupation in ['farmer','student']</code>
State	<code>state == 'GJ' level == 'central'</code>
Category	<code>category in ['SC','ST','OBC']</code>
Education	<code>education == 'undergraduate'</code>
Caste (optional)	Only if user consents to share
Disability	<code>isDisabled == true</code>
Minority	<code>isMinority == true</code>
Farmer	<code>isFarmer == true</code>
Business Owner	<code>isBusinessOwner == true</code>

Example eligibility rule (scheme document)

```
{
  "eligibility": {
    "minAge": 18,
    "maxAge": 40,
    "maxIncome": 250000,
    "gender": ["female"],
    "states": ["ALL"],
    "categories": ["SC", "ST", "OBC", "GENERAL"],
    "requiresFarmer": false,
    "requiresDisability": false
  }
}
```

Matching pseudocode

```
bool isEligible(User u, Scheme s) {
    final e = s.eligibility;
    if (u.age < e.minAge || u.age > e.maxAge) return false;
    if (u.income > e.maxIncome) return false;
    if (e.gender.isNotEmpty && !e.gender.contains(u.gender)) return false;
    if (!e.states.contains('ALL') && !e.states.contains(u.state)) return false;
    if (e.categories.isNotEmpty && !e.categories.contains(u.category)) return false;
    if (e.requiresFarmer && !u.isFarmer) return false;
    if (e.requiresDisability && !u.isDisabled) return false;
    if (e.requiresMinority && !u.isMinority) return false;
    if (e.requiresBusinessOwner && !u.isBusinessOwner) return false;
    return true;
}
// Rank: state-specific first, then nearest deadline, then category match.
```

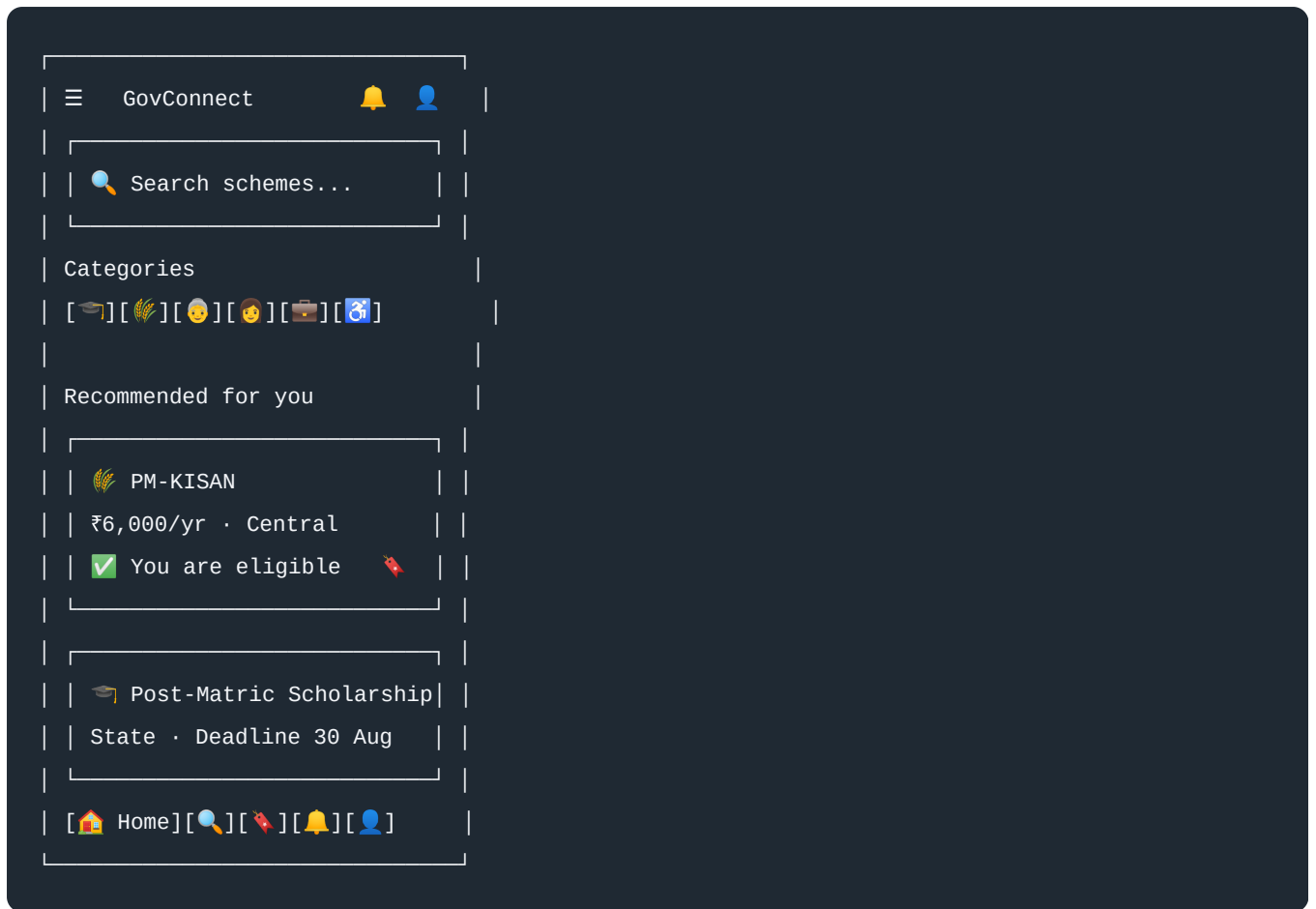
✅ **Why rule-based (not ML) for v1:** explainable ("you qualify because income < ₹2.5L"), no training data needed, easy for admins to verify, and legally safer for government data.

13. UI Pages

Page	Purpose
Splash	Branding + load config/session; route to language or home
Language Selection	Pick UI language on first launch
Login / Guest	Phone OTP auth or continue as guest
Eligibility Quiz	Collect profile answers to power recommendations
Home	Personalized feed of eligible schemes + categories + search entry
Search	Keyword search across schemes
Filters	Refine by state, category, benefit type, target group
Scheme Detail	Full scheme info: benefits, eligibility, documents, deadline, apply link
Bookmarks	Saved schemes for the logged-in user
Notifications	Deadline & new-scheme alerts
Profile	View/edit eligibility answers, language, notification prefs
Settings	Language, notifications, theme, about, logout
Admin Panel	Manage schemes, states, categories, eligibility rules, feedback (web/admin build)

14. Wireframes (Low-Fidelity)

Home



Eligibility Quiz



Scheme Detail

←

PM-KISAN

📌

🔗

🌾

Agriculture · Central

✅

You are eligible

Benefits

- ₹6,000/year in 3 instalments

Eligibility

- Small/marginal farmer

Documents

- Aadhaar · Land record · Bank

Deadline: Open

🌐

Apply on Official Portal

Admin Panel (web)

GovConnect Admin

Schemes | States | Feed

+ New

Search schemes...

Title	Level	State	Active	Edit
PM-KISAN	Central	—	✅	 
Post-Matric Sch	State	GJ	✅	 
Old Age Pension	State	MH	✅	 

15. Color Palette

Government-inspired (trust, India tricolor cues), modern, and WCAG-accessible.

Role	Name	Hex	Usage
Primary	Ashoka Blue	#0B5FA5	App bar, primary buttons, links
Secondary	Deep Saffron	#F4811F	Highlights, active states, CTAs
Accent	India Green	#138808	"Eligible" badges, positive accents
Success	Success Green	#2E7D32	Success messages, confirmations
Warning	Amber	#F9A825	Deadlines approaching, cautions
Error	Error Red	#C62828	Errors, ineligible/invalid
Neutral Dark	Ink	#1F2933	Primary text
Neutral Mid	Slate	#616E7C	Secondary text
Neutral Light	Cloud	#F5F7FA	Backgrounds, cards
White	Surface	#FFFFFF	Surfaces

 Ensure text/background pairs meet **WCAG AA (4.5:1)**. Do not rely on color alone for the "eligible" state — pair with an icon + label.

16. Typography & UI Kit

Fonts (Google Fonts)

Use	Font	Notes
Headings	Poppins	Modern, friendly, strong hierarchy
Body	Inter	Highly legible at small sizes
Indic scripts	Noto Sans (Devanagari/Gujarati/Tamil...)	Full regional-language coverage

Font Hierarchy

Token	Size / Weight	Usage
H1	28 / Bold	Page titles
H2	22 / SemiBold	Section headers
H3	18 / SemiBold	Card titles
Body	16 / Regular	Primary content
Caption	13 / Regular	Meta, labels
Button	16 / SemiBold	CTAs

Spacing

- Base unit **8px**; scale: 4 / 8 / 16 / 24 / 32.
- Screen padding **16px**; card padding **16px**; min tap target **48×48px**.

Icons

- **Material Symbols** (rounded) for consistency; category icons as emoji/custom SVGs.

Buttons

Type	Style
Primary	Filled Ashoka Blue, white text, radius 12
Secondary	Outlined blue, blue text
CTA (Apply)	Filled Saffron, white text
Text/Ghost	Blue text only

Cards

- Radius **16px**, subtle shadow (elevation 1–2), 16px padding, eligibility badge top-right, bookmark toggle inline.

17. Roadmap

Phase	Focus	Key Deliverables	Status
Phase 1	 Research	Scheme data model, eligibility rules research, competitor analysis, user interviews	☐
Phase 2	 Database	Firestore schema, seed 50–100 verified schemes, states & categories, admin panel	☐
Phase 3	 Frontend	Flutter UI (all MVP screens), i18n, auth, navigation	☐
Phase 4	 Recommendation Engine	Rule engine, questionnaire, personalized feed, ranking	☐
Phase 5	 Testing	Unit + widget tests, eligibility accuracy QA, accessibility & device testing, beta with real users	☐
Phase 6	 Deployment	Play Store release, analytics, crash reporting, feedback loop, data-maintenance process	☐

Phase checklist

- ☐ Phase 1 — Research complete
- ☐ Phase 2 — Database & admin ready
- ☐ Phase 3 — Frontend MVP built
- ☐ Phase 4 — Recommendation engine live
- ☐ Phase 5 — Tested & beta validated
- ☐ Phase 6 — Deployed to Play Store

18. Risks

Risk	Impact	Mitigation
Incorrect scheme data	Users misled about eligibility → loss of trust	Verification workflow, updatedAt per scheme, feedback/report button, source citations
Changing government rules	Schemes/eligibility go stale	Scheduled re-verification, admin alerts, versioned rules
Data maintenance burden	Manual curation doesn't scale	Structured admin tools, part-time curators, move toward official APIs
Scalability	High reads spike costs/latency	Firestore indexing, pagination, caching, cost monitoring
User trust	Users doubt a non-government app	"Unofficial, info only" disclaimer, link to official portals, transparency on eligibility reasons
Privacy/legal (sensitive fields)	Caste/disability data is sensitive	Explicit consent, minimal storage, DPDP compliance
Language quality	Poor translations reduce usability	Professional/native review of regional content

19. Future Scope

- ☐  **AI Chatbot** — conversational scheme discovery & Q&A
- ☐  **Voice Assistant** — voice navigation for low-literacy/elderly users
- ☐  **Regional Languages** — expand to 10+ Indian languages
- ☐  **OCR Document Checker** — scan Aadhaar/income proof to auto-verify readiness
- ☐  **Eligibility Prediction** — ML-assisted likely-eligible suggestions
- ☐  **Scheme Deadline Alerts** — smart reminders + calendar sync
- ☐  **Family Profile** — manage benefits for multiple household members
- ☐  **NGO Integration** — partner NGOs assist users with applications

20. Monetization

Core citizen access stays **free**. Monetization targets institutions, not vulnerable users.

Model	Description
 Premium Alerts	Optional paid tier for advanced reminders, priority notifications, document tracking
 Government Partnerships	Paid integrations/sponsored placement for verified official programs
 CSR Sponsorship	Corporates fund the app as a CSR/digital-inclusion initiative
 Analytics Dashboard	Aggregated, anonymized scheme-interest insights for policymakers
 NGO Dashboard	Paid tools for NGOs to help beneficiaries at scale
 White-Label Version	License the platform to state governments / institutions under their brand

✅ **Ready for development.** This document defines product scope, users, data model, algorithm, UI, and roadmap. Next step: **Phase 1 — Research** (verify scheme data sources & finalize the eligibility rule schema), then seed the Firestore `schemes` collection via the Admin Panel.

⚠️ *Statistics in §2 are illustrative placeholders — cite verified sources (MyScheme, NITI Aayog, NSSO) before any external/investor use.*